

Bangladesh University of Business and Technology
Department of Computer Science and Engineering

CSE 498A: Capstone Project Thesis

PROPOSAL

A Explainable AI Federated Learning Framework for Real-Time Continuous Bangla Sign Language to Speech Translation

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1. Introduction

1.1 Problem Statement

Bangla Sign Language (BdSL) is the primary mode of communication for a substantial deaf and hard-of-hearing population in Bangladesh, yet automated interpretation

research remains dominated by isolated recognition of static alphabets, digits, and single words [1]. At least one prior study explicitly identifies continuous, sentence-level recognition as unaddressed future work [2]. Current systems also lack a privacy-preserving training pathway: because sign language video is inherently identity-revealing, centralized collection of raw signer footage raises the same data protection concerns that motivate federated approaches in other sensitive domains such as healthcare [3]. Furthermore, misrecognitions in a deployed interpretation system remain difficult to diagnose or trust without an explainability layer [4], and existing BdSL systems uniformly stop at text output rather than producing spoken audio.

1.2 Motivation

Recent advances in Federated Learning (FL) and transformer-based sequence modeling offer a path to build accurate continuous sign language recognizers without centralizing sensitive signer video [3],[7]. This project is motivated by the need for a complete, privacy-preserving pipeline that not only recognizes continuous BdSL but also makes its decisions interpretable and ultimately speaks the translated result aloud - closing the communication loop between a signer and a hearing listener without requiring a human interpreter to be physically present.

1.3 Overview

This project proposes a federated, explainable framework built on the IsharaKhorob continuous BdSL corpus [5], which pairs signed video with Bangla sentence annotations. The system integrates a transformer-based recognition module trained under a federated protocol, an attention-based explainability layer, a gloss-to-Bangla-text translation stage, and a neural text-to-speech module trained on Bangla speech corpora [6] to produce spoken output. By keeping raw signer video local to each simulated client and aggregating only model weights, the framework mirrors privacy-preserving designs proven in other healthcare and sensitive-data domains while addressing a communication accessibility problem specific to Bangladesh.

2. Objectives

This project develops a privacy-preserving, explainable federated learning framework for continuous Bangla Sign Language recognition and speech translation. It also evaluates the system across recognition accuracy, translation quality, real-time feasibility, and explanation quality.

- **Privacy-Preserving Recognition:** Develop a federated learning (FL) protocol so that continuous BdSL video used to train the recognition model remains local to each simulated client during training.

- Continuous, Transformer-Based Recognition: Implement a CTC-based transformer recognition model that converts unsegmented, sentence-level BdSL video into a gloss sequence, extending beyond landmark-only features with raw-video CNN features.
- Gloss-to-Speech Pipeline: Translate the recognized gloss sequence into grammatical Bangla text and synthesize it into spoken Bangla audio using a neural text-to-speech module.
- Explainable Recognition: Expose and quantitatively evaluate transformer attention weights to visualize which video regions drive each recognition decision.
- System Validation: Evaluate the framework's translation quality, federated accuracy trade-off, real-time latency, and explanation consistency against the closest existing benchmark.

3. Literature Review

Camgöz et al. (2020) [7] proposed Sign Language Transformers, a joint end-to-end architecture for continuous sign language recognition and translation trained with a CTC loss. The study achieved state-of-the-art performance on the RWTH-PHOENIX-2014T benchmark and removed the need for frame-level gloss alignment. However, the model was developed and validated only for German Sign Language and has not been evaluated on low-resource languages such as Bangla.

Yin and Read (2020) [8] proposed the STMC-Transformer to improve gloss-to-text and video-to-text translation quality in continuous sign language translation. The study reported substantial BLEU score gains over prior sequence-to-sequence baselines. However, the approach still depends on the availability of large, gloss-annotated training corpora that do not yet exist at scale for Bangla.

Shin et al. (2023) [9] proposed a combined CNN-transformer architecture for Korean Sign Language recognition. The study demonstrated that hybrid local (CNN) and long-range (transformer) feature extraction is effective for a lower-resource sign language. However, the model was evaluated only on isolated signs rather than continuous sentence-level recognition.

Podder et al. (2022) [10] proposed a deep learning model for Bangla Sign Language alphabet and numeral classification. The study achieved 99.99% accuracy on a large custom image dataset using ResNet18. However, the work is restricted to static,

isolated character and digit recognition and does not address continuous signing or speech output.

Hasan et al. (2025) [11] proposed a multimodal deep learning fusion framework for Bangla Sign Language recognition, combining pose estimation and audio features across a 200-word vocabulary. The study achieved 94.71% accuracy with a late-fusion ViT and VGGish model. However, the approach remains isolated-word rather than continuous, and produces no speech output.

Rubaiyeat et al. (2025) [5] proposed IsharaKhorbor, a 5,642-sentence continuous BdSL corpus scraped from Bangla television news, and benchmarked two gloss-free transformer models (SLTT and a custom RQE-GASLT embedding). The study established that continuous transformer-based BdSL translation is feasible, reporting a best BLEU-4 of 3.88. However, the approach is restricted to landmark-based (MediaPipe pose/hand) features rather than raw video, and does not address federated training, explainability, or speech synthesis.

Kowsher et al. (2022) [12] proposed Bangla-BERT, a monolingual transformer language model for Bangla natural language understanding. The study showed that Bangla-BERT outperforms multilingual alternatives on downstream Bangla NLP tasks. However, the model has not previously been applied as a gloss-to-text translation backbone in a sign language pipeline.

Pfitzner et al. (2021) [3] proposed a systematic review of Federated Learning applications in medical contexts, synthesizing privacy-preserving training designs across healthcare domains. The review highlighted FedAvg as the dominant aggregation protocol for identity-sensitive data. However, the review found no application of federated learning to sign language or gesture-based recognition tasks.

Lamaakal et al. (2026) [15] proposed TinyMSLR, an explainable, edge-deployable hybrid CNN-transformer model for sign language recognition using adaptive fusion and knowledge distillation. The study achieved strong accuracy on German and Mandarin isolated-sign benchmarks while remaining lightweight enough for edge devices. However, the work addresses isolated-sign classification only, does not involve federated training, and was not evaluated on Bangla Sign Language.

4. Methodology

The proposed methodology presents a privacy-preserving, explainable federated learning framework for continuous Bangla Sign Language recognition and speech

translation. It includes data preprocessing, federated distributed training, attention-based explainability, gloss-to-text translation, and neural speech synthesis.

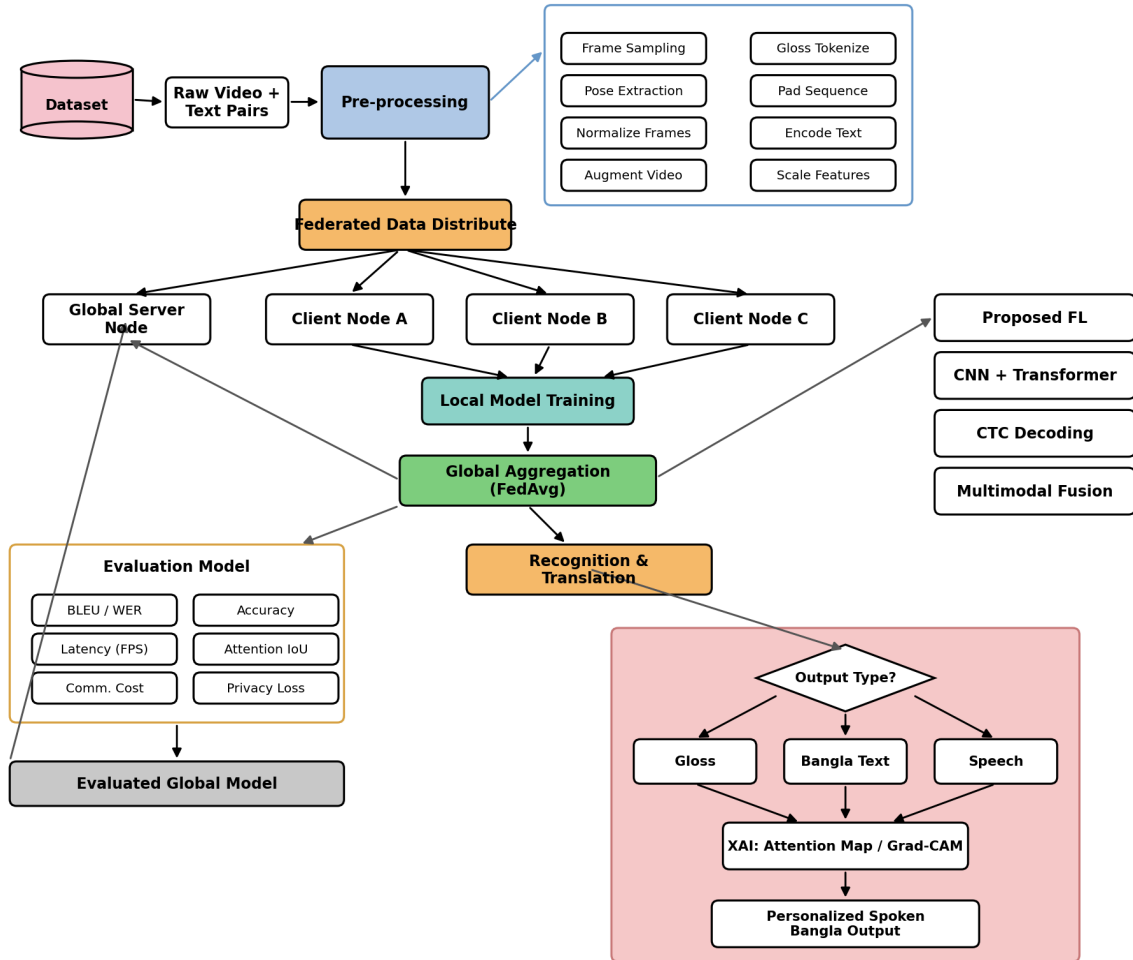


Figure 1: Proposed Federated Learning Framework for Continuous BdSL to Speech Translation.

The proposed framework follows a structured approach represented in the architecture above:

- **Data Preprocessing:** Extracting video frames and pose keypoints from the IsharaKhorbor corpus, and tokenizing, padding, and encoding the paired Bangla gloss and text annotations.
- **Federated Client Partitioning:** Distributing the preprocessed corpus across simulated client nodes (grouped by signer) so that raw video never leaves its originating client.

- **Federated Recognition Training:** Training a CNN-transformer recognition model with CTC decoding locally at each client, then aggregating model weights at the global server using the FedAvg algorithm.
- **Explainable Recognition:** Extracting transformer attention weights from the aggregated global model to visualize and quantitatively evaluate which video regions drove each gloss prediction.
- **Translation and Speech Synthesis:** Converting the recognized gloss sequence into grammatical Bangla text using a Bangla-BERT-based translation stage, then synthesizing the text into spoken Bangla audio using a neural text-to-speech model.

4.1 Federated Training Configuration

Each client trains locally for a fixed number of local epochs on its own partition before sending a model update to the server, rather than triggering updates by a raw data-volume threshold. To keep each client's local training statistically meaningful, every simulated client is allocated at least 500–700 sentences (IsharaKhobor's 8 signers are grouped into 3–4 clients, keeping each above this minimum); a client with too little data risks overfitting and produces a noisy, unreliable update. The following configuration will be used as a starting point, to be tuned empirically:

Parameter	Value
<i>Local epochs per round</i>	2–3
<i>Batch size</i>	16 or 32
<i>Communication rounds</i>	20–30 (with early stopping on validation loss plateau)
<i>Aggregation algorithm</i>	FedAvg
<i>Minimum data per client</i>	500–700 sentences

4.2 Demonstration Interface

To validate the framework end-to-end and support project defense, a lightweight web-based demonstration interface will be built (using Streamlit or Gradio) allowing a user to upload or record a short continuous BdSL video clip and view the recognized gloss sequence, translated Bangla text, an attention-map visualization, and play back the synthesized speech output. This interface is scoped as a functional prototype for demonstration purposes rather than a production deployment, keeping engineering effort focused on the core recognition, federated training, and explainability research contributions.

5. Expected Outcomes

- A functional, privacy-preserving federated learning framework will be developed for continuous Bangla Sign Language recognition.
- Improved recognition and translation quality will be demonstrated relative to the existing landmark-only IsharaKhobor baseline (BLEU-4 of 3.88), through the addition of raw-video features.
- A secure, decentralized training pipeline will be established, keeping raw signer video local to each simulated client throughout training.
- An explainable recognition layer will be delivered, allowing attention maps to be inspected for trust and error diagnosis.
- An end-to-end spoken-output system will be produced, extending BdSL interpretation beyond text for the first time in the reviewed literature.

6. Complex Engineering Problem Mapping

Mapping with Knowledge Profile (WK):

- WK2 (Mathematics): Application of Federated Averaging (FedAvg), CTC sequence loss, and BLEU/WER statistical evaluation.
- WK3 (Engineering Fundamentals): Implementation of decentralized system architectures, transformer sequence modeling, and multimodal (video-text-audio) data fusion.
- WK4 (Specialist Knowledge): Expertise in Computer Vision, Natural Language Processing (NLP), Speech Synthesis, and Federated Learning.

Mapping with Complex Engineering Problems (WP):

- WP1 (Depth of knowledge): Requires a deep understanding of continuous sequence recognition, decentralized AI training, and sensitive video data handling.
- WP2 (Conflicting requirements): Balancing the trade-off between recognition accuracy and strict data-locality privacy constraints imposed by the federated protocol.
- WP6 (Stakeholder involvement): Involves deaf and hard-of-hearing signers, hearing listeners, schools for the deaf, and interpreters, each with distinct privacy and usability needs.

Mapping with Complex Engineering Activities (EA):

- EA1 (Range of resources): Uses GPU-enabled computing, PyTorch/Flower federated learning frameworks, and the IsharaKhorbor and OpenSLR datasets.
- EA3 (Innovation): Applies federated and explainable AI, previously unexplored for any sign language, to a communication accessibility problem in Bangladesh.
- EA4 (Consequences to society): Directly supports the deaf and hard-of-hearing community by enabling communication with hearing individuals without requiring a human interpreter.

7. References

- [1] S. Das, M. S. Imtiaz, N. H. Neom, et al., "A hybrid approach for Bangla sign language recognition using deep transfer learning model with random forest classifier," *Expert Systems with Applications*, vol. 213, 118914, 2022. <https://doi.org/10.1016/j.eswa.2022.118914>
- [2] T. Tazalli, Z. A. Aunshu, S. S. Liya, et al., "Computer Vision-Based Bengali Sign Language To Text Generation," in *Proc. IPAS* 2022. <https://doi.org/10.1109/ipas55744.2022.10052928>
- [3] B. Pfitzner, N. Steckhan, and B. Arnrich, "Federated Learning in a Medical Context: A Systematic Literature Review," *ACM Transactions on Internet Technology*, vol. 21, no. 2, 2021. <https://doi.org/10.1145/3412357>
- [4] R. Guidotti, A. Monreale, S. Ruggieri, et al., "A survey of methods for explaining black box models," *ACM Computing Surveys*, vol. 51, no. 5, 2019. <https://doi.org/10.1145/3236009>
- [5] H. A. Rubaiyeat, H. Mahmud, and M. K. Hasan, "Bangla Sign Language Translation: Dataset Creation, Challenges, Benchmarking and Prospects," *arXiv:2511.21533*, 2025. <https://arxiv.org/abs/2511.21533>
- [6] A. Ahmad, M. Selim, and M. Z. Iqbal, "SUST TTS Corpus: A phonetically-balanced corpus for Bangla text-to-speech synthesis," *Acoustical Science and Technology*, vol. 42, no. 6, 2021. <https://doi.org/10.1250/ast.42.326>
- [7] N. C. Camgöz, O. Koller, S. Hadfield, and R. Bowden, "Sign Language Transformers: Joint End-to-End Sign Language Recognition and Translation," in *Proc. CVPR* 2020. <https://doi.org/10.1109/cvpr42600.2020.01004>

- [8] K. Yin and J. Read, "Better Sign Language Translation with STMC-Transformer," in Proc. COLING 2020. <https://doi.org/10.18653/v1/2020.coling-main.525>
- [9] J. Shin, A. S. M. Miah, M. A. M. Hasan, et al., "Korean Sign Language Recognition Using Transformer-Based Deep Neural Network," *Applied Sciences*, vol. 13, no. 5, 3029, 2023. <https://doi.org/10.3390/app13053029>
- [10] K. K. Podder, M. E. H. Chowdhury, A. Tahir, et al., "Bangla Sign Language (BdSL) Alphabets and Numerals Classification Using a Deep Learning Model," *Sensors*, vol. 22, no. 2, 574, 2022. <https://doi.org/10.3390/s22020574>
- [11] A. Hasan, M. H. Jobayer, M. Pias, et al., "Bangla Sign Language Recognition With Multimodal Deep Learning Fusion," *Engineering Reports*, vol. 7, 2025. <https://doi.org/10.1002/eng2.70139>
- [12] M. Kowsher, A. A. Sami, N. J. Prottasha, et al., "Bangla-BERT: Transformer-Based Efficient Model for Transfer Learning and Language Understanding," *IEEE Access*, vol. 10, 2022. <https://doi.org/10.1109/access.2022.3197662>
- [13] J. C. Jiang, B. Kantarci, S. Oktuğ, et al., "Federated Learning in Smart City Sensing: Challenges and Opportunities," *Sensors*, vol. 20, no. 21, 6230, 2020. <https://doi.org/10.3390/s20216230>
- [14] A. Holzinger, A. Saranti, C. Molnar, et al., "Explainable AI Methods - A Brief Overview," in *xxAI - Beyond Explainable AI*, LNCS, 2022.
- [15] I. Lamaakal, et al., "TinyMSLR: an explainable hybrid CNN-transformer model for sign language recognition on edge devices using adaptive fusion and knowledge distillation," *Scientific Reports*, 2026. <https://doi.org/10.1038/s41598-026-38478-8>